

Outline of Frequency domain analysis of CT signals



THE HONG KONG
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AND TECHNOLOGY

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7 Fourier Transform of Periodic Signals

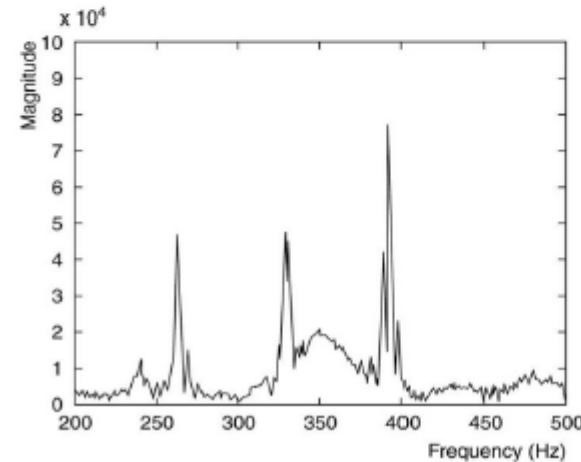
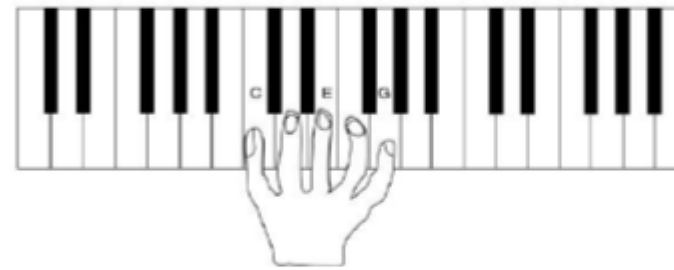
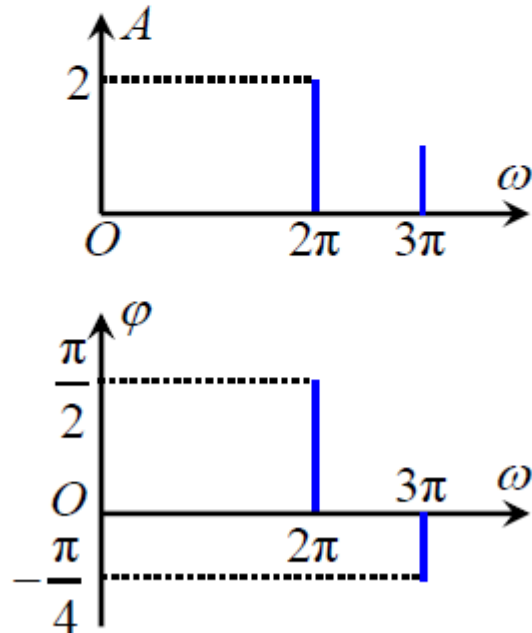
8 Energy Spectrum and Power Spectrum

1. Introduction to the Spectrum Concept

$$f(t) = 2 \cos\left(2\pi t + \frac{\pi}{2}\right) + \cos\left(3\pi t - \frac{\pi}{4}\right)$$

**Frequency
Domain Analysis:
More Intuitive**

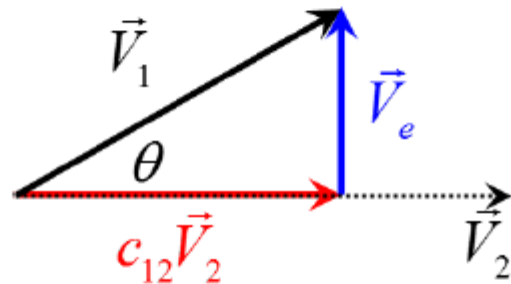
Spectrum Diagram:



2. Orthogonal Decomposition of Signals

Vector Orthogonal Decomposition

$$\vec{V}_1 = c_{12}\vec{V}_2 + \vec{V}_e$$

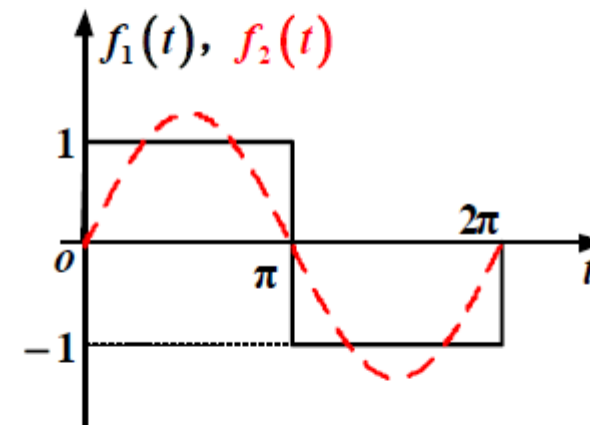


$$c_{12} = \frac{\vec{V}_1 \cdot \vec{V}_2}{\vec{V}_2 \cdot \vec{V}_2}$$

Coefficient Selection Principle:
Minimize the error component

Signal Orthogonal Decomposition

$$f_1(t) = c_{12}f_2(t) + f_e(t)$$



$$c_{12} = \frac{\int_{t_1}^{t_2} f_1(t) f_2^*(t) dt}{\int_{t_1}^{t_2} f_2(t) f_2^*(t) dt}$$

3. Complete Orthogonal Function Set

Decomposing signals using an orthogonal function set

$$f(t) = \sum_{r=1}^n c_r g_r(t) + f_e(t)$$

$$= c_1 g_1(t) + c_2 g_2(t) + \dots + c_n g_n(t) + f_e(t)$$

$g_1(t), g_2(t), \dots, g_n(t)$ form an **orthogonal function set**

Projection coefficients under the principle of minimum mean square error

$$c_r = \frac{\int_{t_1}^{t_2} f(t) g_r^*(t) dt}{\int_{t_1}^{t_2} g_r(t) g_r^*(t) dt}$$

3. Complete Orthogonal Function Set

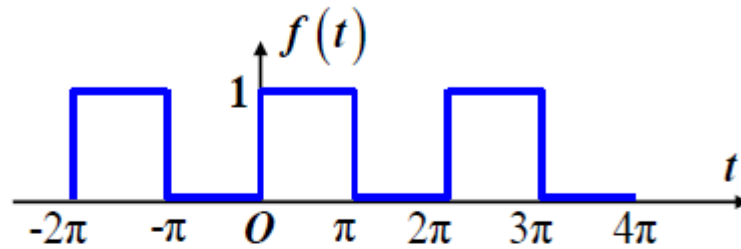
$$\begin{aligned} f(t) &= \sum_{r=1}^n c_r g_r(t) + f_e(t) \\ &= c_1 g_1(t) + c_2 g_2(t) + \dots + c_n g_n(t) + f_e(t) \end{aligned}$$

$$n \rightarrow +\infty \Rightarrow \overline{f_e^2(t)} \rightarrow 0$$

Complete Orthogonal Function Set

No function exists outside the set that is orthogonal to all functions within the set

Example 1



Expand into trigonometric
Fourier series

Solution:

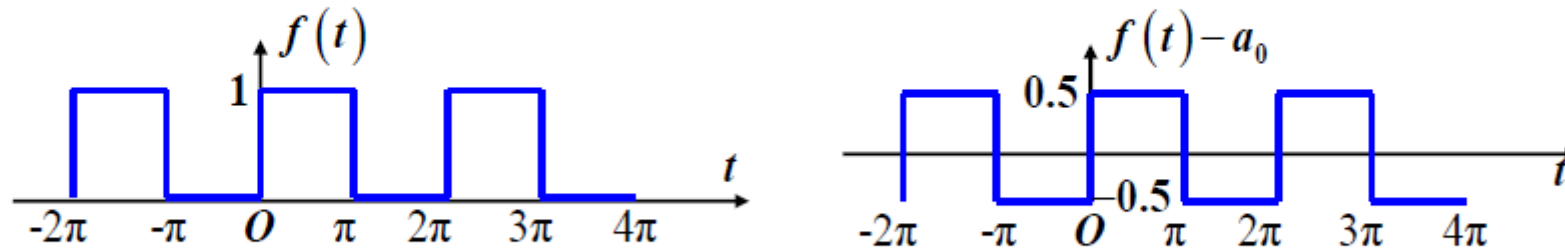
The **trigonometric form** Fourier series of a signal can be expressed as:

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \right]$$

Signal period $T_0 = 2\pi$, fundamental angular frequency:

$$\omega_0 = \frac{2\pi}{T_0} = 1$$

Example 1



Find the DC component coefficient a_0

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(t) dt = \frac{1}{2}$$

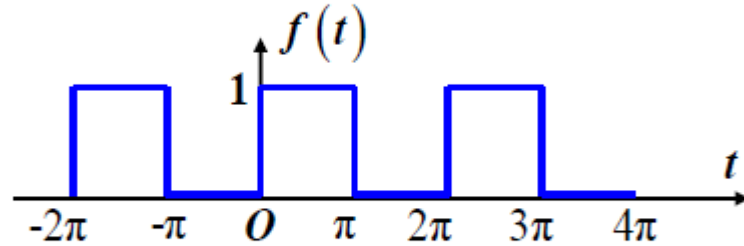
$f(t) - a_0$ is an odd function, and also an odd harmonic function

Cosine component coefficient a_n

$$a_n = 0$$

Example 1

Find the sine component coefficient b_n

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(t) \sin(nt) dt$$

$$= \frac{1}{\pi} \int_0^{\pi} \sin(nt) dt = \frac{1 - \cos(n\pi)}{n\pi}$$

The Fourier series can be expressed as

$$f(t) = \frac{1}{2} + \sum_{n=1}^{\infty} \left[\frac{1 - \cos(n\pi)}{n\pi} \cdot \sin(nt) \right]$$
$$= \frac{1}{2} + \frac{2}{\pi} \sin(t) + \frac{2}{3\pi} \sin(3t) + \frac{2}{5\pi} \sin(5t) \dots$$

Example 2

Given $f(t) = 1 + \sqrt{5} \sin \omega_0 t - \cos \left(2\omega_0 t - \frac{3\pi}{4} \right),$

Plot its magnitude spectrum and phase spectrum

Solution:

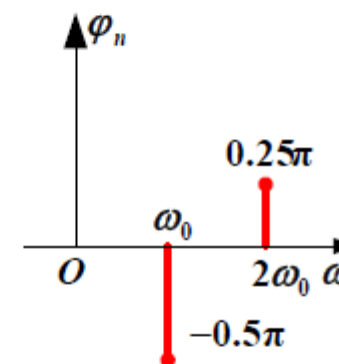
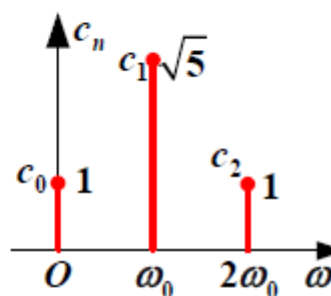
Convert to harmonic form, expressed in cosine function form:

$$\begin{aligned} f(t) &= 1 + \sqrt{5} \cos \left(\omega_0 t - \frac{\pi}{2} \right) + \cos \left(2\omega_0 t - \frac{3\pi}{4} + \pi \right) \\ &= 1 + \sqrt{5} \cos \left(\omega_0 t - \frac{\pi}{2} \right) + \cos \left(2\omega_0 t + \frac{\pi}{4} \right) \end{aligned}$$

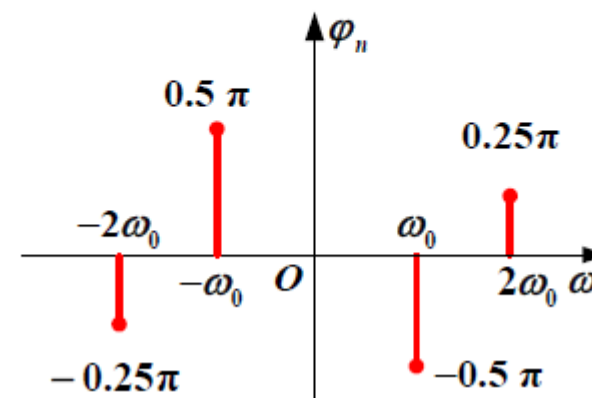
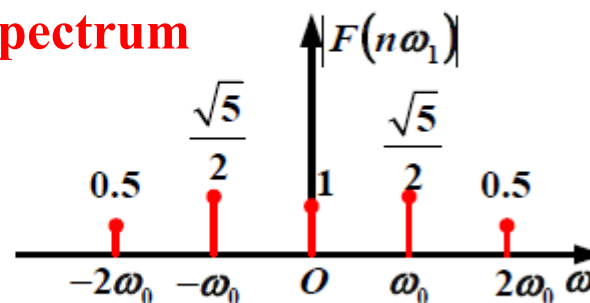
Example 2

$$f(t) = 1 + \sqrt{5} \cos\left(\omega_0 t - \frac{\pi}{2}\right) + \cos\left(2\omega_0 t + \frac{\pi}{4}\right)$$

One-Sided Spectrum



Double-Sided Spectrum



6. Basic Properties of Fourier Transform

Example 4

Find the Fourier transform of the sampling signal $\text{Sa}(\omega_c t)$

Analysis

Using the Fourier transform definition:

$$F_1(\omega) = \int_{-\infty}^{\infty} \text{Sa}(\omega_c t) e^{-j\omega t} dt$$

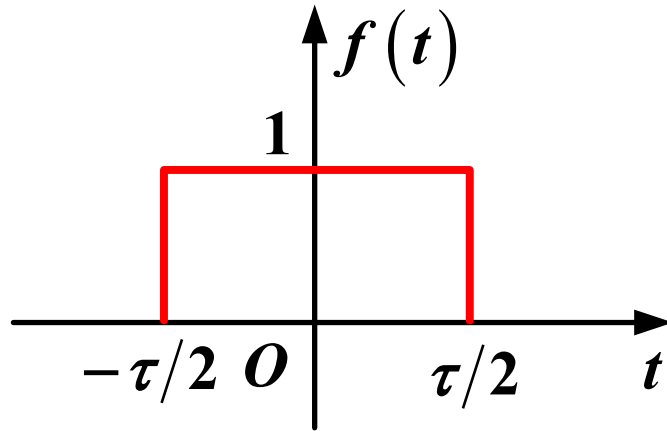
$$\int_{-\infty}^{\infty} \text{Sa}(\omega_c t) e^{-j\omega t} dt$$

This integral is challenging to solve directly

Therefore, we attempt to solve it using **duality property**

6. Basic Properties of Fourier Transform

Example 4

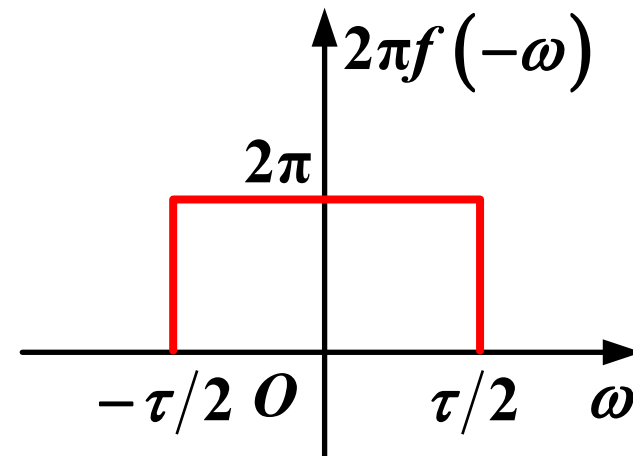


$$\Leftrightarrow F(\omega) = \tau \text{Sa}\left(\frac{\tau}{2}\omega\right)$$

Obtain a new
transform pair using
duality property

$$F(t) = \tau \text{Sa}\left(\frac{\tau}{2}t\right)$$

$\Leftrightarrow$

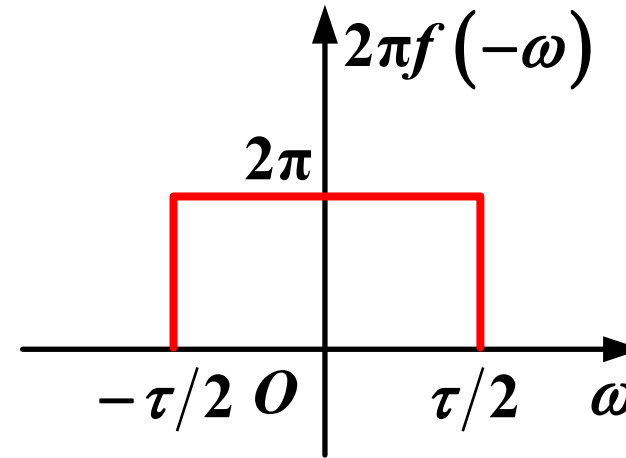


6. Basic Properties of Fourier Transform



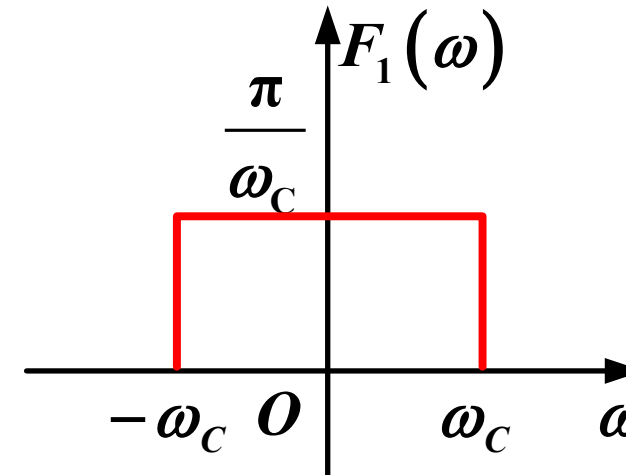
Example 4

$$F(t) = \tau \text{Sa}\left(\frac{\tau}{2}t\right) \quad \leftrightarrow$$

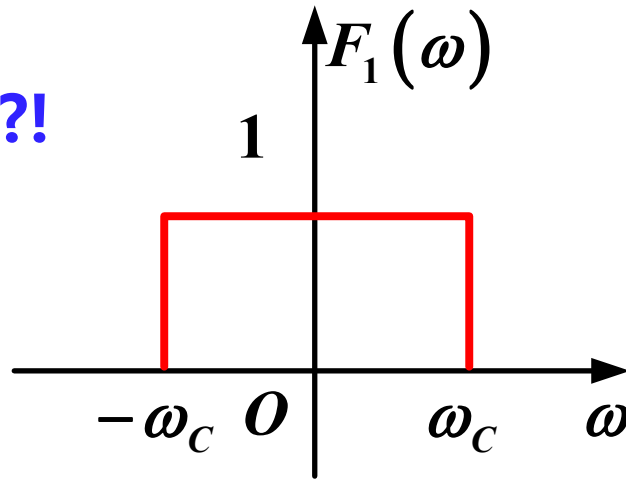


Let $\frac{\tau}{2} = \omega_c$, divide both sides by $2\omega_c$

$$\text{Sa}(\omega_c t) \quad \leftrightarrow$$



Which Circuit ?!



Time domain function ?